Mechanics 1

Week 7

Shear stress

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Learning outcome

Types of stresses

Shear stress

Video recap

$$\therefore \quad \tau = F\frac{A\bar{y}}{Ib}$$

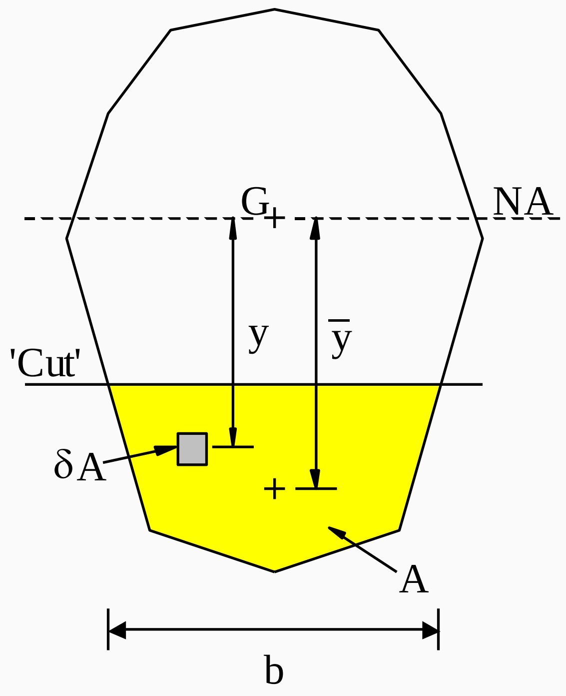
where,

F = total vertical shear force on the cross-section

I = second moment of area of the cross-section about the neutral axis

b = breadth of the cross-section at the point under consideration

$A\bar{y}$ = first moment of area about the neutral axis of the portion of the cross-section 'beyond' the point.



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Example 1

A rolled steel I-section has properties:

Overall depth = 450 mm

Flange thickness = 12 mm

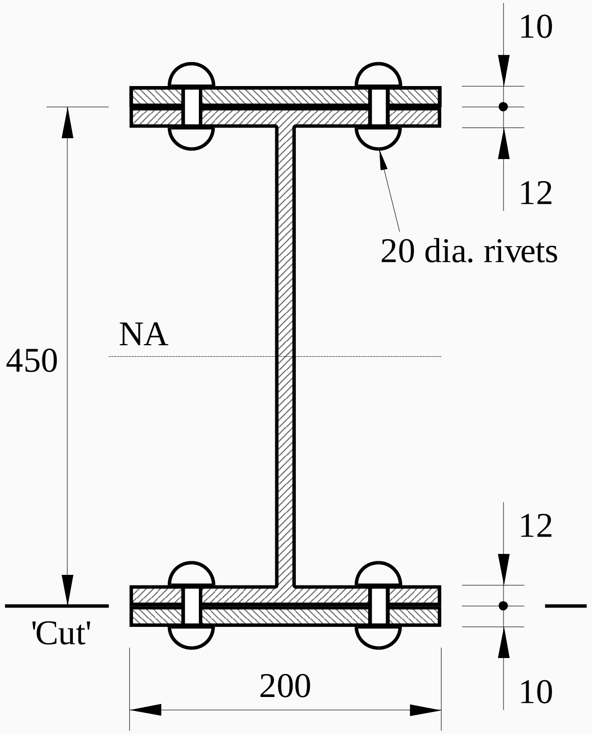
(Gross) second moment of area I = 3.0×10^8 mm^4

The section has been reinforced by riveting 200 x 10 mm steel plates to the upper and lower flanges, using 20 mm diameter rivets. The pitch (longitudinal spacing) of the rivets is 160 mm.

What is the shear force acting on each rivet due to a total vertical shear force of 180 kN in the beam?

$$I = (3.0\times10^8) + 2 \times ((1/12) \times (200 \times 10^3) + (200 \times 10) \times 230^2)$$
$$- 4 \times ((1/12) \times (20 \times 22^3) + (20 \times 22) \times 224^2)$$
$$= 4.233 \times 10^8 \text{ mm}^4$$

Area beyond cut, A = (200 x 10) – 2(20 x 10) = 1600 mm^2



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Shear flow, q = $\dfrac{FA\bar{y}}{I} = \dfrac{(180x10^3)x1600x230}{4.233x10^8} = 156.48\ N/mm$

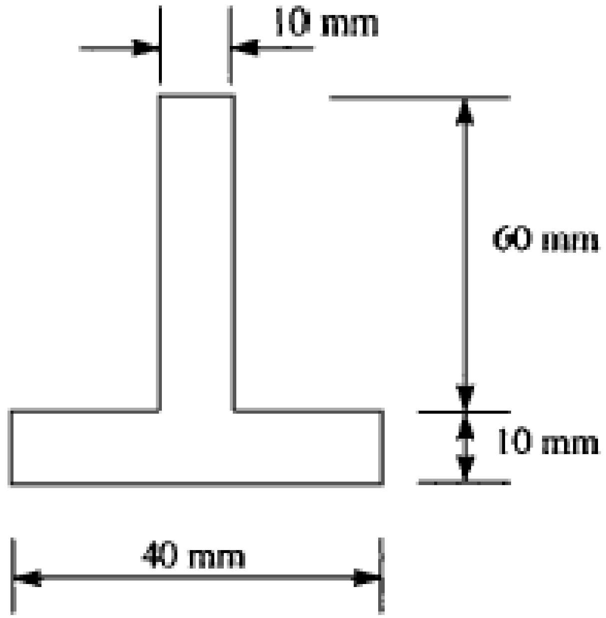
Number of rivets per unit length, n = 2/160 = 0.0125 rivets / mm

Shear force per rivet = q/n = 156.48 / 0.0125 = 12.52 kN

Example 2

A cantilever has the inverted T-section shown below. It carries a vertical shear load of **4 kN** in a downward direction. Determine the distribution of vertical shear stress in its cross-section.

Find Neutral Axis and Second moment of Area



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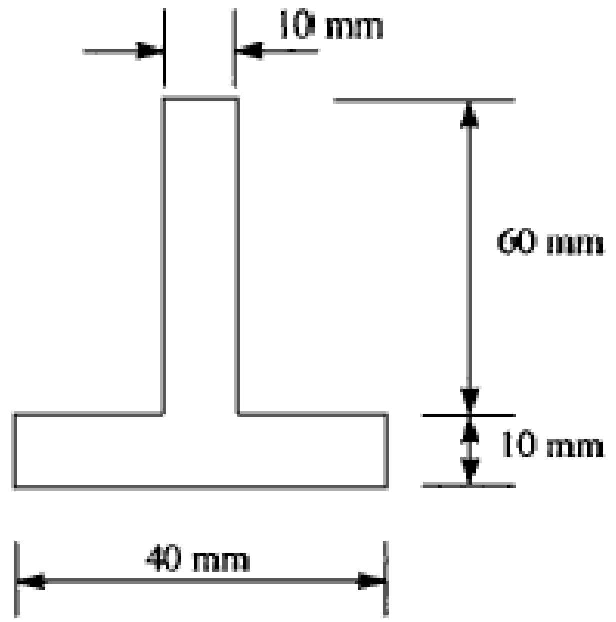
1. Calculate the location of the NA of the shape.

$$(40x10 + 60x10)\,\overline{y_1} = 10x60x30 + 40x10x65$$

$$\overline{y_1} = 44mm$$



2. Calculate I for the whole shape.

$$I = \frac{10x60^3}{12} + 600(30 - 44)^2 + \frac{40x10^3}{12} + 400(65 - 44)^2$$

$$I = 4.773x10^5 mm^4$$

3. Consider a ‘cut’ in the web

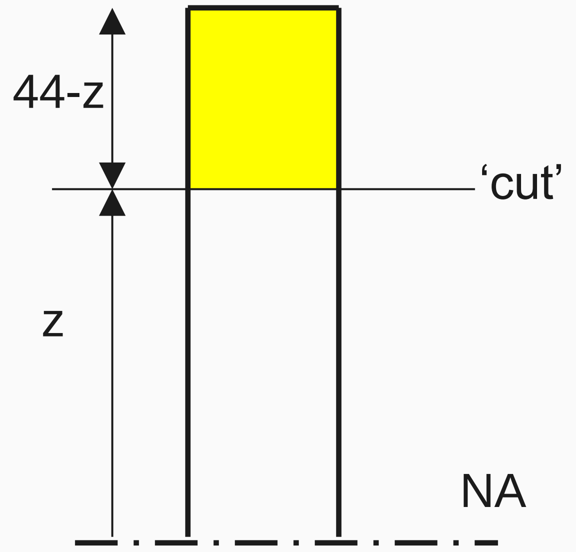
Area beyond cut = 10(44 - z)

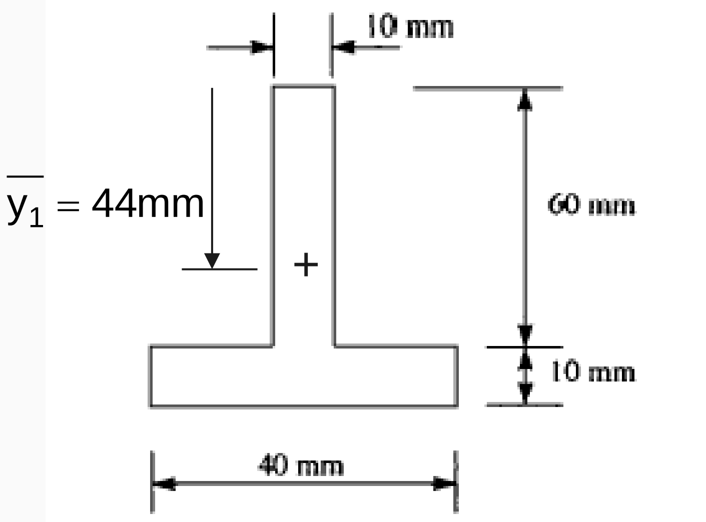
$$\overline{y_2} = \frac{44 - z}{2} + z = \frac{44 + z}{2}$$

$$A\overline{y_2} = 5\left(44^2 - z^2\right)$$

$$\tau = F\frac{A\overline{y}}{Ib} = \left(4x10^3\right)\frac{5\left(44^2 - z^2\right)}{4.73x10^5x10}$$

$$\tau = 0.004\left(44^2 - z^2\right) \qquad N/mm^2$$





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4. Consider a 'cut' in the flange

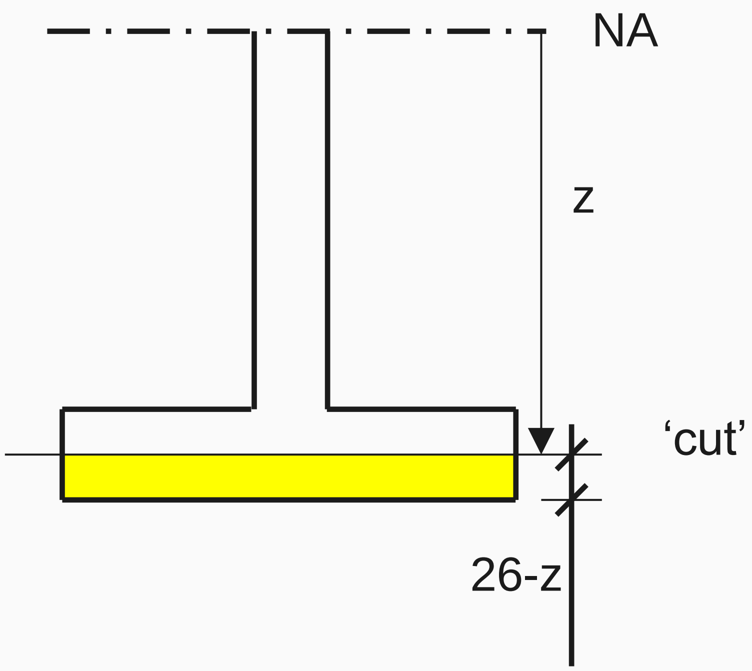
Area beyond cut = 40(26-z)

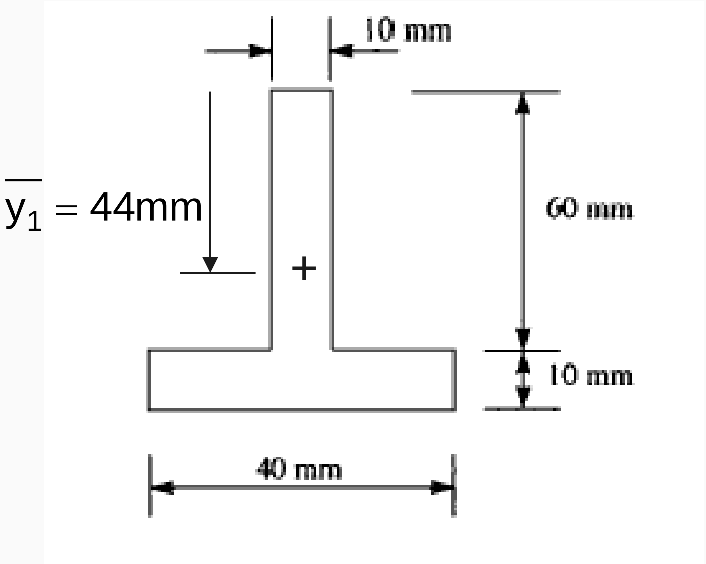
$$\overline{y_3} = \frac{26 - z}{2} + z = \frac{26 + z}{2}$$

$$A\overline{y_3} = 20\left(26^2 - z^2\right)$$

$$\tau = 4x10^3 \frac{20\left(26^2 - z^2\right)}{4.73x10^5 x40}$$

$$\tau = 0.004\left(26^2 - z^2\right) \quad N/mm^2$$





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3 equations

$$\tau = 0.004\left(44^2 - z^2\right) \quad N/mm^2$$

Web (above NA)

$$\tau = a\left(16^2 - z^2\right) \quad N/mm^2$$

Web (below NA)

$$\tau = 0.004\left(26^2 - z^2\right) \quad N/mm^2$$

Flange)

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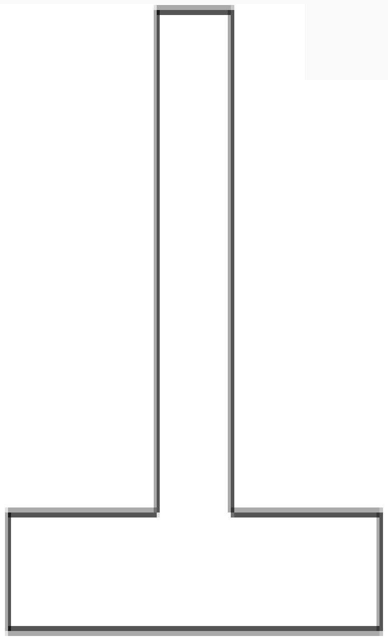
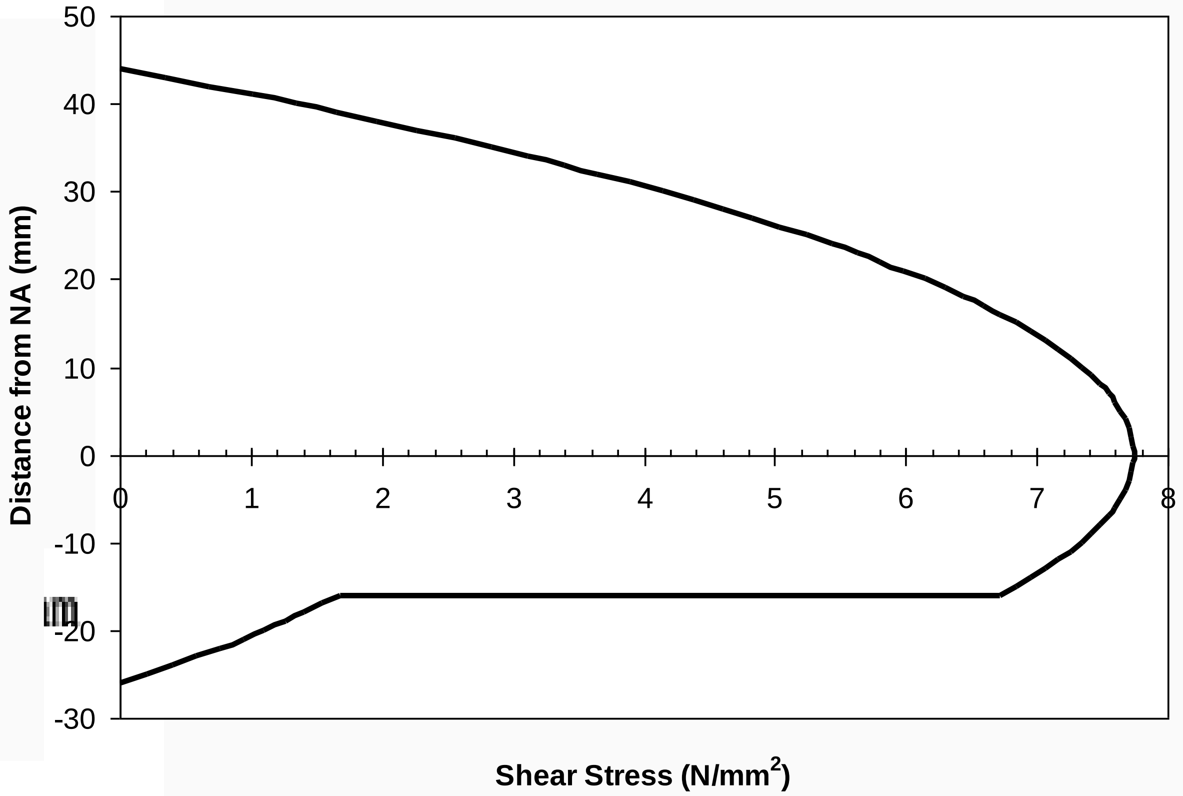



Distance from NA (mm)
50
40
30
20
10
0
-10
-20
-30
0
1
2
3
4
5
6
7
8
Shear Stress (N/mm²)

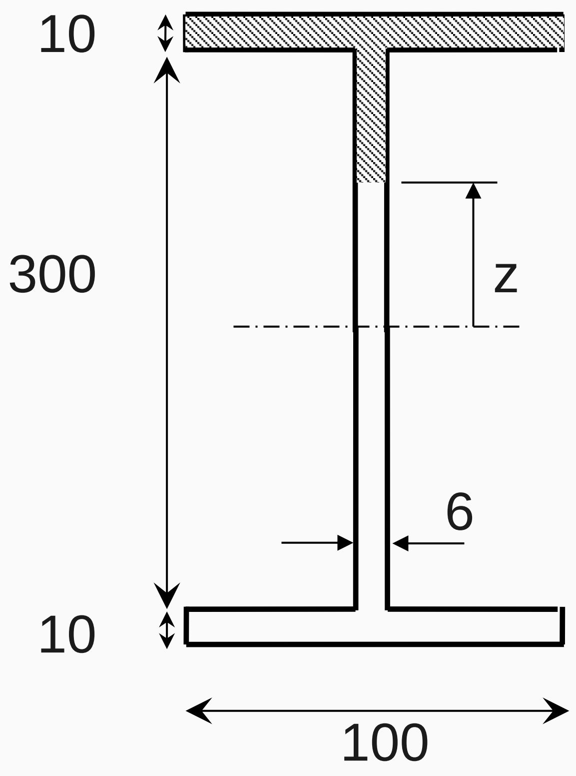
Example 3

The I beam shown below is subjected to a shear force of 50kN. Derive an expression for the shear stress in the web and hence calculate the shear stress at the NA and at the web/flange interface.

$$\tau = \frac{FA\overline{y}}{Ib}$$

$I = (6 \times 300^3)/12 + 2(100 \times 10^3/12 + 100 \times 10 \times 155^2)$

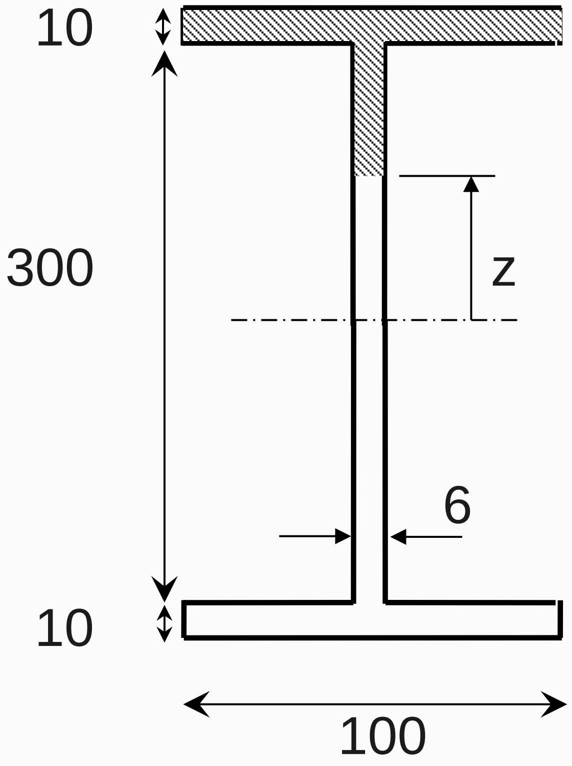
$I = 61.57 \times 10^6 \text{ mm}^4$



Example 3

$$A\bar{y} = \left[(100x10)x155 + \left(6x(150 - z)x\frac{150 + z}{2}\right)\right]$$

$$A\bar{y} = 155000 + 3\left(150^2 - z^2\right)$$



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Example 3

$$\tau = F\frac{A\bar{y}}{Ib} = \frac{(50x10^3)}{(61.57x10^6)x6}\left[155000 + 3\left(150^2 - z^2\right)\right]$$

$$\tau = 135.35x10^{-6}\left[155000 + 3\left(150^2 - z^2\right)\right]$$

When z = 0, τ = 30.12 N/mm^2

When z = 150, τ = 20.98 N/mm^2